Pavel Andreevich Rozhkov

C++ Software Engineer — HPC & GPU Computing

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Summary

C++ software engineer specializing in high-performance computing (HPC), parallel programming, and GPU acceleration with CUDA. Experienced in developing and optimizing computational algorithms, designing complex software architectures, and improving CPU/GPU performance. Skilled in working with large C++ codebases, with a strong mathematical foundation from the Faculty of Mechanics and Mathematics at Moscow State University.

Work Experience

Rock Flow Dynamics June 2023 – Present

Senior C++/CUDA Developer *Moscow*

- Develop and optimize a high-performance numerical simulation engine in C++20 and CUDA for CPUs, GPUs, and distributed computing systems.
- Profile and optimize computational algorithms, memory management, CPU/GPU data transfers, and GPU matrix assembly.
- Develop multithreaded and distributed algorithms using C++, MPI, and CUDA, ensuring reproducible numerical results.
- Extend and refactor a large C++ codebase, conduct code reviews, and perform testing and performance analysis.

Achievements:

- Reduced total runtime for large batches of simulations by 10%, lowering cluster resource consumption.
- Reduced GPU memory usage by 20–25% by optimizing data storage structures.
- Developed an optimized numerically stable summation algorithm for multithreaded computations, reducing overhead from approximately 10% to less than 1%.
- Implemented numerically stable summation for MPI and improved the reproducibility of parallel computations across several simulator subsystems.
- Achieved a fivefold speedup in CPU/GPU data transfers before Newton iterations.
- Accelerated the GPU matrix assembly component by a factor of 6.
- Optimized numerical model initialization, reducing one of the most expensive components' share of total simulation runtime from 22% to 2% on a model with performance issues.
- Accelerated the reporting parameter calculation module by more than 100 times by developing and applying a more efficient algorithm.
- Developed a segmentation system for zonal modeling of well construction and integrated the design module with other simulation subsystems.
- Developed a well trajectory optimization algorithm that accelerated calculations by more than 20 times on workloads involving hundreds of models.

Education

Lomonosov Moscow State University Moscow

Faculty of Mechanics and Mathematics, Specialist Degree in Mathematics *2020 – 2027 (in progress)*

Skills

Languages: C/C++23, Python, Go, SQL
HPC & GPU: CUDA C++, MPI, OpenMP, multithreading, distributed computing, numerical computing, CPU/GPU optimization, memory management
Libraries: STL, Boost, GoogleTest, Qt, OpenCV, zlib
Profiling & Debugging: Nsight Compute, Nsight Systems, perf, GDB, CUDA-GDB, Valgrind
Tools & Platforms: Linux, Git, GN/Ninja, CMake, GCC, Clang, Jenkins
English: B2